2

(4) ✓ $\int \sin^3\theta \, d\theta$

$= \int \sin^2\theta \cdot \sin\theta \, d\theta$

$= \int (1-\cos^2\theta)\sin\theta \, d\theta$

$= \int \sin\theta \, d\theta - \int \sin\theta\cos^2\theta \, d\theta$

$= -\cos\theta + \frac{1}{3}\cos^3\theta + C$

3

(5) ✓ $\int \frac{dx}{3-5\sin 2x}$

$= -\int \frac{1}{5\sin 2x - 3} \, dx$

$u = 2x$

$\frac{du}{dx} = 2$

$= -\frac{1}{2}\int \frac{1}{5\sin u - 3} \, du$

⋮

$= \frac{1}{8} \ln \left| \frac{\tan x - 3}{3\tan x - 1} \right| //$

8. 2

1

(4) V $\int \tan 5\theta \sec^4 5\theta \, d\theta$

$\tan 5\theta = t$

$5\sec^2 5\theta \, d\theta = dt$

$= \int \tan 5\theta \, (1+\tan^2 5\theta) \sec^2 5\theta \, d\theta$

$= \int \frac{1}{5} t \, (1+t^2) \, dt$

$= \frac{1}{20} t^4 + \frac{1}{10} t^2 + C$

$= \frac{1}{20} \tan^4 5\theta + \frac{1}{10} \tan^2 5\theta + C$ //

(12) V $\int \csc^4 \frac{3}{2}\theta \, d\theta$

$(\csc x)' = -\csc \cot$

$\sec' x = \sec \tan$

$\cot' x = -\csc^2 x$

$= \int (1+\cot^2 \frac{3}{2}\theta) \csc^2 \frac{3}{2}\theta \, d\theta$

$= \int \csc^2 \frac{3}{2}\theta \, d\theta + \int \cot^2 \frac{3}{2}\theta \cdot \csc^2 \frac{3}{2}\theta \, d\theta$ // t

$\cot \frac{3}{2}\theta = t$

$-\frac{3}{2} \csc^2 \frac{3}{2}\theta \, d\theta = dt$

$= -\frac{2}{3} \cot \frac{3}{2}\theta + \int -\frac{2}{3} t^2 \, dt$

$= -\frac{2}{3} \cot \frac{3}{2}\theta - \frac{2}{9} \cot^3 \frac{3}{2}\theta + C$ //

(14) V $\int \cot^2 2y \csc^4 2y \, dy$

$\cot 2y = t$

$-2\csc^2 2y \, dy = dt$

$\int \cot^2 2y \cdot \csc^2 2y \, (1+\cot^2 2y) \, dy$

$= \int \cot^2 2y \csc^2 2y + \cot^4 2y \csc^2 2y \, dy$

$= \int -\frac{1}{2} t^2 \, dt + \int -\frac{1}{2} t^4 \, dt$

$= -\frac{1}{6} t^3 - \frac{1}{10} t^5 + C$

$= -\frac{1}{6} \cot^3 2y - \frac{1}{10} \cot^5 2y + C$ //

2

(2) V $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cot x \csc^3 x \, dx$

$\csc x = t$

$-\csc x \cot x \, dx = dt$

$-\int_2^1 t^2 \, dt = \int_1^2 t^2 \, dt = \left[\frac{1}{3} t^3\right]_1^2 = \frac{8}{3} - \frac{1}{3} = \frac{7}{3}$ //